MATH 205

# Linear Algebra

## LECTURE 2 · INVERSES, TRANSPOSE & ELEMENTARY MATRICES

*Properties worth trusting, a new operation, and the machinery that actually computes an inverse.*

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WELCOME BACK

# Where Today Fits

## Last time

- Linear equations, systems, and the augmented matrix
- Matrix addition, scalar multiplication, equality
- Matrix multiplication — and why $AB \neq BA$
- The identity matrix, and the definition of $A^{-1}$

## Today

- Properties inverses always satisfy
- A new operation: the transpose
- How to actually compute $A^{-1}$

## Course Roadmap

1. Systems of Linear Equations
2. **Matrices & Matrix Algebra**
3. Determinants
4. Vector Spaces
5. Eigenvalues & Eigenvectors

MOTIVATION

# Why Inverses & Transpose Matter

*The same objects show up everywhere once you know to look for them.*

### Cryptography

The Hill cipher encrypts blocks of text by multiplying by a matrix — decrypting is just multiplying by its inverse.

### Economics

Leontief's input-output model predicts production levels by solving $(I - C)^{-1}d$ — a matrix inverse at the center of a Nobel-winning model.

### Computer Graphics

To undo a camera transform or un-rotate an object, you apply the inverse of the matrix that moved it.

### Statistics & ML

Covariance matrices are always symmetric — $A^T = A$ shows up the moment you compute variances between variables.

### Numerical Computing

MATLAB, NumPy's linalg.inv and every solver under the hood row-reduce with elementary matrices — exactly today's method.

### Engineering Design

Solve Ax = b for many different loads b by reusing one pre-computed $A^{-1}$ instead of re-solving from scratch each time.

QUICK RECAP

# The Inverse Matrix

Where we left off last time:

$$QA = AQ = I_n = I \qquad \Longleftrightarrow \qquad Q = A^{-1}$$

**Nonsingular / invertible** — an inverse of A exists. **Singular** — no inverse exists.

*Today we find out this definition immediately buys us more than it looks like — starting with a question you might not have thought to ask: could a matrix have two different inverses?*

# Theorem: The Inverse Is Unique

If B and C are both inverses of the matrix A, then **B = C**.

### Proof

∵ B is an inverse of A, we have BA = I.

Multiplying both sides on the right by C:

$$
\begin{aligned}
&BA = I && (1)\\
&(BA)C = IC = C && (2)\\
&(BA)C = B(AC) = B(I) = B && (3) \quad (\because AC = I)\\
&\therefore\ C = B && \text{from } (2) \text{ and } (3)
\end{aligned}
$$

# Theorem: Inverse of a Product

If A and B are invertible matrices of the same size, then

$$(AB)^{-1} = B^{-1}A^{-1}$$

***Notice the order flips*** *— this matters, since matrix multiplication doesn't commute.*

**Proof — verify $B^{-1}A^{-1}$ satisfies the definition of an inverse for AB:**

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I \quad (1)$$

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I \quad (2)$$

*Both products equal I, so $B^{-1}A^{-1}$ is indeed the inverse of AB.*

# In Short

1. A product of any number of invertible matrices is invertible.

2. The inverse of the product is the product of the inverses, in reverse order.

If $A_1$, $A_2$, …, $A_n$ are invertible matrices of the same size:

$$(A_1 A_2 \cdots A_n)^{-1} = A_n^{-1} A_{n-1}^{-1} \cdots A_2^{-1} A_1^{-1}$$

*Reverse the order every single time you invert a product — it's the one rule everyone forgets first.*

# Powers of a Matrix

Notation: for a square matrix A and integer n > 0,

$$A^n = \underbrace{A \cdot A \cdots A}_{n \text{ times}} \qquad (n > 0)$$

## Try It Yourself

(1) If A is invertible:

$$(A^{-1})^{-1} = A$$

**Proof:**

$$\because \; AA^{-1} = A^{-1}A = I \quad \Longrightarrow \quad (A^{-1})^{-1} = A$$

(2) Prove:

$$(A^n)^{-1} = (A^{-1})^n \qquad \text{for } n = 0, 1, 2, \ldots$$

*Hint:*

*write $A^n$ = AA ... A (n times), and pair each factor with its inverse.*

# A New Operation: The Transpose

**Transpose:** if A is an m × n matrix, the transpose of A (denoted $A^T$) is obtained by interchanging the rows and columns of A. The order of $A^T$ is n × m.

**Example**

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}_{2\times 3} \qquad A^T = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \\ a_{13} & a_{23} \end{bmatrix}_{3\times 2}$$

*The first row of A becomes the first column of $A^T$, and so on.*

# Two Properties of the Transpose

### Double Transpose

$$(A^T)^T = A$$

Interchanging rows and columns twice leaves a matrix unchanged — transposing is its own inverse operation.

### Transpose of a Sum

$$(A+B)^T = A^T + B^T$$

*if A and B are the same size. Check it with general 2×2 entries:*

$$(A+B)^T = \begin{bmatrix} a_{11}+b_{11} & a_{21}+b_{21} \\ a_{12}+b_{12} & a_{22}+b_{22} \end{bmatrix} = A^T + B^T$$

# Symmetric Matrices

A square matrix A is called **symmetric** if $A^T = A$

$$I^T = I \quad \text{(always)}$$

Identity is always symmetric

$$\begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}^T = \begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}$$

Every diagonal matrix is symmetric

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 4 & 3 \\ 2 & 3 & 5 \end{bmatrix}$$

General example: $a_{12}=a_{21}=1$, $a_{13}=a_{31}=2$, $a_{23}=a_{32}=3$

# A Useful Note

While taking the transpose, the **diagonal entries** of a square matrix never change position.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \qquad A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}$$

# Skew-Symmetric Matrices

A square matrix A is called **skew-symmetric** if $A^T = -A$

$$B = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 3 \\ -2 & -3 & 0 \end{bmatrix}, \qquad B^T = \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & -3 \\ 2 & 3 & 0 \end{bmatrix} = -B$$

Diagonal entries in a skew-symmetric matrix are always zero. Why?

$$a_{11} = -a_{11} \implies a_{11} = 0, \text{ etc.}$$

*Since $a_{ii}$ can only equal its own negative when $a_{ii} = 0$.*

TRY IT YOURSELF

# In-Class Practice — 3 Minutes

Compute: For the matrix below:

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & -2 & 0 \end{bmatrix}, \quad \text{find } A^2 \text{ and } A^3$$

**Hints**

- $A^2 = A \cdot A$ — use the row-times-column recipe from last lecture
- $A^3 = A^2 \cdot A$, reusing the $A^2$ you just found

*This is Assignment 1, Q1(a) — parts (b) and (c) push further: showing $A^3 = A^2 + A - 5I$, then using that identity (no more multiplying!) to find $A^4$ and $A^{-1}$.*

# Back to Systems: A Bonus from Invertibility

If $A\mathbf{x} = \mathbf{b}$ represents a system of n equations in n unknowns, and A is invertible, the solution is unique.

**Proof**

Let $x_1$, $x_2$ both be solutions:

$$\begin{aligned} & A\mathbf{x}_1 = \mathbf{b}, \quad A\mathbf{x}_2 = \mathbf{b} && \Rightarrow \; A\mathbf{x}_1 = A\mathbf{x}_2 \\ \Rightarrow \; & A^{-1}(A\mathbf{x}_1) = A^{-1}(A\mathbf{x}_2) && (\because A \text{ is invertible}) \\ \Rightarrow \; & I\mathbf{x}_1 = I\mathbf{x}_2 && \Rightarrow \; \mathbf{x}_1 = \mathbf{x}_2 \end{aligned}$$

$\therefore$ $A^{-1}$ is unique, and $\mathbf{x} = A^{-1}\mathbf{b}$ is the unique solution.

THE QUESTION

# How Do We Actually Find $A^{-1}$?

*Everything so far assumed $A^{-1}$ exists. Time to actually compute it — starting from the same row operations you already use to solve systems.*

# Elementary Row Operations

*Three moves, applied to a matrix — the same three you already use to solve systems by hand.*

Starting matrix:

$$B = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 1 & 4 & 4 & 0 \end{bmatrix}$$

**Multiply a row by a nonzero constant**

$R_2 \to 2R_2$

$$\begin{bmatrix} 1 & 0 & 2 & 3 \\ 4 & -2 & 6 & 12 \\ 1 & 4 & 4 & 0 \end{bmatrix}$$

**Interchange two rows**

$R_1 \leftrightarrow R_2$

$$\begin{bmatrix} 2 & -1 & 3 & 6 \\ 1 & 0 & 2 & 3 \\ 1 & 4 & 4 & 0 \end{bmatrix}$$

**Add a multiple of one row to another**

$R_1 \to R_1 + 2R_2$

$$\begin{bmatrix} 5 & -2 & 8 & 15 \\ 2 & -1 & 3 & 6 \\ 1 & 4 & 4 & 0 \end{bmatrix}$$

# Equivalent Matrices

If A is obtained from B by one or more elementary row operations, A is called **row equivalent** (or just equivalent) to B, and vice versa — written **A ~ B** or **B ~ A**.

From the previous slide's three operations, all applied to the same starting matrix B:

- B ~  (row-scaled matrix)
- B ~ (row-swapped matrix)
- B ~ (row-combined matrix)

*Row equivalence is the relationship every step of Gaussian elimination preserves — you never leave the equivalence class of your starting matrix.*

# Elementary Matrices

**Elementary matrix:** an n × n matrix obtained from the n × n identity matrix $I_n$ by performing a single elementary row operation.

**Example**

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$$

$$R_3 \to R_3 + 3R_1$$

E is obtained from $I_3$ by performing exactly one operation on the identity: adding 3 × (row 1) to row 3.

*Every elementary matrix is invertible — its inverse just undoes the one operation that created it.*

# The Key Result

Let's apply E to our earlier example — watch what happens compared to the row operation done by hand:

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 1 & 4 & 4 \end{bmatrix}, \qquad E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$$

Consider EA — multiply E and A directly:

$$EA = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 4 & 10 \end{bmatrix} = B$$

**An elementary row operation on a matrix has the same effect as premultiplying by the elementary matrix that corresponds to that same operation.**

RECAP

# Key Takeaways

### Inverses are unique

if B and C both invert A, then B = C — there's only ever one $A^{-1}$.

### Order reverses

$(AB)^{-1} = B^{-1}A^{-1}$, and this extends to any product of invertible matrices.

### The transpose

flips rows and columns; $(A^T)^T = A$, and $(A+B)^T = A^T + B^T$.

### Symmetric & skew-symmetric

$A^T = A$ vs. $A^T = -A$ — the second forces every diagonal entry to be zero.

### Unique solutions

when A is invertible, $Ax = b$ has exactly one solution: $x = A^{-1}b$.

### Elementary matrices

encode row operations — premultiplying by E does exactly what the row operation does.

NEXT TIME

# Looking Ahead

Lecture 3

## Finding $A^{-1}$ by Row Reduction

- The recipe: [A | I] → [I | $A^{-1}$]
- Why it works: chaining elementary matrices
- Solving Ax = b entirely via $A^{-1}$
- A first look at determinants

### Homework

**See CANVAS & OneNote**

**Assignment 1**

### Reflection

*The reverse-order rule for $(AB)^{-1}$ trips up almost everyone once — bring a 2×2 example if you want to see it fail in real time.*

# Questions?

*Inverses, Transpose & Elementary Matrices · Lecture 2 complete*

**NEXT: FINDING $A^{-1}$ BY ROW REDUCTION**